

CHEN SHAO

cshao676@gmail.com

EDUCATION

PhD Candidate in Computer Science

July 2022 – Present

Karlsruhe Institute of Technology (KIT), Germany

Supervised by Prof. York Sure-Vetter and Dr. Tobias Käfer (Web of Science in AIFB)

M.Sc. in Electronic and Information Engineering

Oct. 2018 – Mar. 2022

Karlsruhe Institute of Technology (KIT), Germany

Grade: 2.1

Thesis: “Energy Difference Prediction in Chemistry Using Graph Neural Networks”

Supervised by Prof. Pascal Friederich (AiMat, ITI)

B.Eng. in Electronic and Information Engineering

Sept. 2013 – June 2017

Xi'an Jiaotong University (XJTU), China

Grade: 1.8

Thesis: Ground-Roll Noise Removal Based on Multiscale Geometric Transformation

Honors: Siyuan Scholarship (Top 5% of class)

Awards: First Prize, National Undergraduate Innovation and Entrepreneurship Competition

TECHNICAL EXPERTISE

NLP: Fine-tuning and pre-training of language models, multimodal alignment.

AI/ML Architecture: Generative Models (Flow Matching/Diffusion), Graph Foundation Models.

Software & Engineering: Python (PyTorch, PyG, JAX, TensorFlow), C/C++, Docker, MATLAB.

Languages: Chinese (Native), English (Business Fluent), German (C1).

PROJECTS

Research Intern, Max Planck Institute of Biochemistry

01/2026 – Present

Foundation Models for Protein Design

Munich, Germany

Supervised by Dr. Dexiong Chen and Prof. Karsten Borgwardt

- Developed and optimized pre-training algorithms for foundation models in molecular property prediction.

Research Associate (Research, PI), Karlsruhe Institute of Technology

06/2022 – Present

Graphs in Time Series and Natural Sciences

Karlsruhe, Germany

LLM-GALP (LLM Graph Reasoning (PI), Funder BMFTR)

09/2024 – Present

- Developed SOTA memory-efficient LLM fine-tuning for text-attributed graphs and submitted to PKDD (Top Journal in Data Mining).

ML4RW (Responsible AI for Energy System (PI), Funder KIT-ARRTI)

01/2024 – 01/2025

- Designed a state-of-the-art approach for time series forecasting and provided a new benchmark of European energy dataset for large-scale, non-stationary time-series research.

KIGLIS (AI in 6G with Nokia Bell Lab, VPI Photonics, Funder BMFTR)

06/2022 – 01/2024

- Actively contributed in a collaboration project *Nokia Bell Labs* and *VPI Photonics*.
- Proposed two novel algorithms for 6G channel equalization and presented them in ECOC and OFC.

Research Assistant (Hiwi), Karlsruhe Institute of Technology (KIT)

11/2018 – 06/2022

Different Engagements in signal processing and machine learning

Karlsruhe, Germany

Autonomous Learning Robots (ALR Lab), KIT
Hiwi Student: Graph Neural Networks for Finite Element Analysis
Supervised by Niklas Freymuth (KIT)

11/2021 – 04/2022

- Generated a synthetic Finite Element Simulation dataset to train an adaptive meshing algorithm, supporting reinforcement learning environments for high-resolution simulators.

Institute of Industrial Information Technology (IIIT), KIT
Hiwi Student: Algorithm Development for Hyperspectral Imaging
Supervised by Dr. Maximilian Schambach (KIT, SAP)

06/2020 – 04/2021

- Developed software to systematically process light field data using time-frequency decomposition.

Precitec GmbH & Co. KG
Industry Research Intern, Deep Learning for Laser Material Processing

Gaggenau/Karlsruhe, Germany

04/2021 – 09/2021

- Benchmarked five deep learning architectures for image classification and optimized performance for laser material processing video sequences.

Xi'an Jiaotong University (XJTU)
Research Intern, Ground-roll Noise Removal Based on Multiscale Geometric Transformation
Supervised by Prof. Wenchao Chen (XJTU)

Xi'an, China

01/2016 – 06/2017

- Developed and adapted Ridgelet algorithms for noise suppression in raw seismic data.

PRINCIPAL INVESTIGATOR IN FUNDED PROJECTS

(ML4RW) Machine Learning-Based Algorithm for Maximizing Resource Waste Reduction.
Funded by *KIT Academy for Responsible Research, Teaching, and Innovation (ARRTI)*, 2024.

EUR 74,000

(LLM-GALP) Large Language Model Featured Graph Algorithms for Link Prediction.
Funded by *Federal Ministry of Education and Research (BMFTR)*, 2025–2026.

EUR 105,800

AWARDS AND FELLOWSHIPS

Software Campus Fellowship – “Werde EntrepreNerd”
Awarded for outstanding publication record and leadership potential.

2025 – 2026

Siyuan Scholarship (Top 5%)
Academic excellence award at Xi'an Jiaotong University.

2016

National Innovation Competition (First Prize)
Recognized for independent cutting edge research at XJTU.

2017

Provincial String Competition (First Prize)
First place in a province-level high school violin competition.

2011

SELECTED PUBLICATIONS & AWARDS

Nature Communications Materials (2022): “Review of Graph Neural Network for Materials”.

TKDD/KDD (2025): Lead author on Graph Automorphism & Link Prediction research.

Awards: DLR Fellowship (2025), Siyuan Scholarship, National Innovation Competition Winner.

PEER-REVIEWED PUBLICATIONS

Zhenyi Zhu, **Chen Shao**, et al. *Deep Switching Koopman Networks: Discovery of Switching Dynamic Regimes in Industrial Systems. Proceedings of the 2026 IEEE 22nd International Conference on Automation Science and Engineering (CASE 2026)*, Shenyang, China, August 17–21, 2026.

Chen Shao, Zhenyi Zhu, and Danai Koutra. *When Forecasting Fails: Overcoming Temporal Correlation Shift in Time Series*. In *ICML 2026 Workshop on Foundation Models for Structured Data (FMSD)*, 2026.

Chen Shao, Joella Wang, Zhanbo Huang, Danai Koutra. *DynGCN: Capturing Dynamic Correlation with Message Passing*. In *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)*, 2026.

Chen Shao, Joella Wang, Zhanbo Huang, Tobias Kaefer, Michael Faerber, Benjamin Schaefer. *BigElectricity: A Benchmark for Scalable Multivariate Time Series Forecasting. Proceedings of the ACM International Conference on Information and Knowledge Management (CIKM)*, 2025.

Chen Shao, Ruirui Wang, Tobias Kaefer, Danai Koutra. *Link Prediction on Text-Attributed Graphs: A New Benchmark and Efficient LM-Nested GNN Design*. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 2026.

Chen Shao, Elias Giacomidis, Shi Li, Jialei Li, Michael Faerber, Tobias Kaefer, and Andre Richter. *A Novel Machine Learning-Based Equalizer for a Downstream 100G PAM-4 PON*. *Optical Fiber Communication Conference and Exhibition (OFC 2024)*, Oral Presentation (Top 5%).

Chen Shao, Elias Giacomidis, Tobias Kaefer, Michael Faerber. *Time-Series Forecasting Methods in Fiber Nonlinearity Mitigation*. *European Conference on Optical Communication (ECOC 2024)*, Oral Presentation (Top 5%).

Chen Shao, Elias Giacomidis, Michael Faerber, Tobias Kaefer. *Time-Series Prediction Methods in Fiber Nonlinearity Mitigation*. *Photonics*, 2024. Special Issue: Machine Learning Applied to Optical Communication Systems (Invited Paper).

Patrick Reiser, **Chen Shao**, Chen Zhou, Andre Eberhard, Houssam Metni, Marlen Neubert, Timo Sommer, Clint van Hoesel, Luca Torresi, Pascal Friederich. *Review of Graph Neural Networks for Materials*. *Communications Materials*, 3, 93 (2022).

PREPRINTS

Chen Shao, Donald Loveland, Danai Koutra. *Edge-Level Automorphism in GNNs: A Quantitative Framework and Effective Designs for Link Prediction*. (Submitted to ICLR27).

Chen Shao, Chen Zhou, Pascal Friederich. *Graph Neural Networks to Learn Joint Representations of Disjoint Molecular Graphs*. *arXiv:2210.09517* (2022).

REFERENCES

Dr. Tobias Käfer

Senior Researcher / Group Leader
Karlsruhe Institute of Technology (KIT)
Institute AIFB
Email: tobias.kaefer@kit.edu
Relationship: Doctoral Supervisor

Prof. Danai Koutra

Associate Professor of Data Science and AI
University of Michigan, Ann Arbor
Computer Science and Engineering
Email: dkoutra@umich.edu
Relationship: Mentor